PROTEA HEIGHTS ACADEMY · IT

Nexus WiFi setup guide

Setting up the Linux server and the Ruckus vSZ for a sign-in page: from an empty server to the first pilot sign-ins with school Google accounts.

IN THIS GUIDE

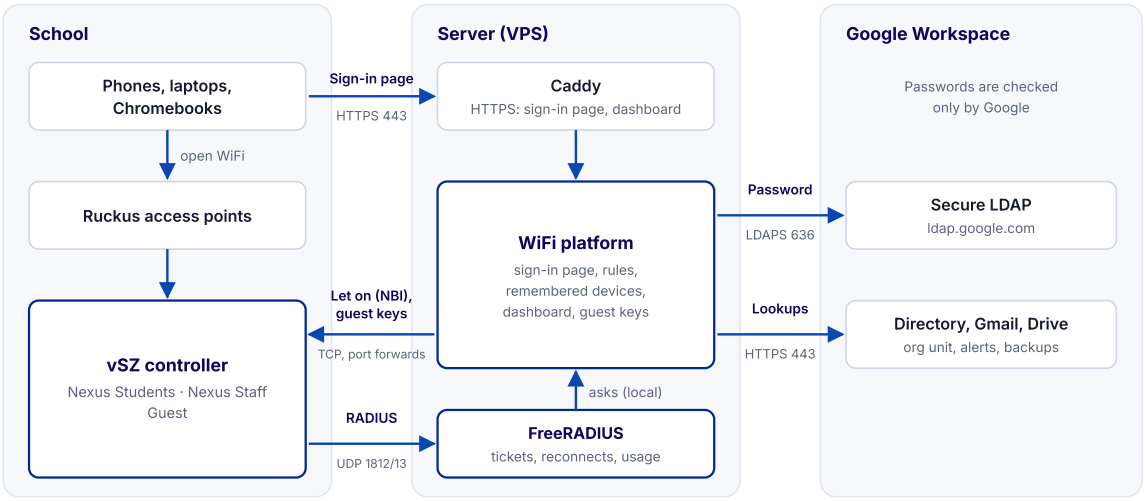
- What to collect and decide before you start
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For	Software	Date	Source
School IT staff	PHA WiFi 0.2.0, sign-in page	5 November 2026	docs/setup-guide.html in the repo

OVERVIEW

How it fits together

People join an open network, **Nexus Students** or **Nexus Staff**. A sign-in page opens by itself, and they type their school Google email and password. Nothing is installed on the device. Google checks the password, the platform checks the person may use that network, and the device is remembered for the rest of the term. Visitors use **Guest** with keys that admins make in the dashboard.



One sign-in, step by step

1. A phone joins **Nexus Students**. The vSZ asks the server whether this device is remembered. If it is, and the person is still allowed, the phone is online at once and nothing below happens.
2. Otherwise the vSZ sends the phone's browser to `https://wifi.phahs.org.za/`, and the phone shows it in its "Log in to network" window.
3. The person types their email and password. The platform checks the password with **Google Secure LDAP**, then checks the domain, blocks, the Google directory (active? which org unit?), the rules, and that this is the right network for them.
4. The platform asks the vSZ to let the phone on, through the vSZ's **Northbound Portal Interface (NBI)**, with a one-time ticket instead of the password. The vSZ checks the ticket with FreeRADIUS, which asks the platform.
5. The phone is online on that network's VLAN, and the platform remembers it for 90 days.

The password goes only to Google, and is never stored or logged. The vSZ and RADIUS only ever see the one-time ticket, which works once, for two minutes.

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PART 1

Plan

Collect these before you start. Filling in the sheet below saves going back and forth between the server, Google and the vSZ.

You'll need

- **A Linux VPS** with a fixed public IP address: Ubuntu Server 24.04 LTS, 1 vCPU, 1 GB RAM and 20 GB of disk is plenty.
- **DNS access** for `phahs.org.za`, to add two records.
- **A super admin account** in the Google Admin console, and access to Google Cloud for the same domain.
- **Admin access to the vSZ** and to the school's firewall.
- **VLANs for staff, learners and guests** that already reach every access point's switch port, each with a DHCP scope and DNS.
- A PC with an SSH client. Windows 10 and 11 include `ssh` and `scp` in PowerShell.

Worksheet

Item	Example	Yours
Server public IP	198.51.100.20	
School public IP: where the vSZ's traffic appears on the internet	203.0.113.10	
Outside port for the vSZ API (forwarded to vSZ 8443)	9443	
Outside port for the vSZ NBI (forwarded to vSZ 9443)	9444	
vSZ software version	6.1.1	
Your own public IP(s), for SSH	203.0.113.10	
Staff network and VLAN	Nexus Staff, VLAN 1	
Learner network and VLAN	Nexus Students, VLAN 10	
Guest network and VLAN	Guest, VLAN 60	
Google org units for staff	/Staff, /IT Staff	
Google org unit for learners	/Students	
How long a device is remembered	90 days (a term)	
Dashboard admins	llotter@phahs.org.za	
Mailbox alerts are sent from	wifi-alerts@phahs.org.za	
People who get alerts	llotter@phahs.org.za	

Secrets never go on this sheet
The RADIUS shared secret, the platform token, the vSZ API and NBI passwords, the Google OAuth client secret, the LDAP access credentials, and the Google key and certificate files go in a password manager, never on paper, in email or in chat. Part 2 shows how to make the first two.

If the server is down

- **No new sign-ins and no reconnects** on Nexus Students and Nexus Staff until the server is back. The sign-in page can't load, and remembered devices can't be checked.
- **Devices already online stay online** until their session ends, at most 12 hours.
- **Guest keeps working.** Guest keys live on the vSZ. Only making new keys needs the server.

PART 2

Prepare the Linux server

This part sets up an Ubuntu 24.04 server with key-only SSH, a firewall, automatic security updates and Docker, then puts the code and the settings folders in place.

2.1 Create the server and its DNS names

1. Create the VPS with **Ubuntu Server 24.04 LTS**. If the provider has its own firewall, allow the same ports as in 2.4.
2. Add two DNS **A records** pointing at the server's public IP:
   - `wifi.phahs.org.za`: the sign-in page and the admin dashboard.
   - `radius.phahs.org.za`: FreeRADIUS needs a certificate for this name to start. Devices never see it.
3. Check both from your PC. Each should print the server's IP:

ON YOUR PC (POWERSHELL)

```
nslookup wifi.phahs.org.za
nslookup radius.phahs.org.za
```

2.2 First login, updates and an admin user

Log in as `root` (or the user your provider made). These examples call the admin user `itadmin`.

ON THE SERVER, AS ROOT

```
apt update && apt full-upgrade -y
timedatectl set-timezone Africa/Johannesburg
apt install -y git unattended-upgrades
adduser itadmin
usermod -aG sudo itadmin
```

Copy your SSH key to the new user. If you don't have a key yet, run `ssh-keygen -t ed25519` on your PC first.

ON YOUR PC (POWERSHELL)

```
type $env:USERPROFILE\.ssh\id_ed25519.pub | ssh itadmin@SERVER_IP "mkdir -p ~/.ssh && cat >> ~/.ssh/authorized_keys &&
chmod 700 ~/.ssh && chmod 600 ~/.ssh/authorized_keys"
```

Log in as `itadmin` with the key. Then turn off password logins and root logins:

ON THE SERVER, AS ITADMIN

```
sudo tee /etc/ssh/sshd_config.d/10-pha.conf >/dev/null <<'EOF'
PasswordAuthentication no
PermitRootLogin no
EOF
sudo systemctl restart ssh
sudo dpkg-reconfigure -plow unattended-upgrades    # answer Yes
```

> **Keep your current session open**
> Before you close it, open a second SSH session as `itadmin` and check it works. If it doesn't, fix it from the first session.

2.3 Install Docker

These are Docker's own instructions for Ubuntu. They install Docker Engine and the Compose plugin.

```
sudo apt-get install -y ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
echo "deb [arch=amd64 signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu noble stable" |
sudo tee /etc/apt/sources.list.d/docker.list
sudo apt-get update
sudo apt-get install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
sudo docker run --rm hello-world       # prints "Hello from Docker!"
sudo docker compose version
```

Type the `echo` line as one line. If `apt-get update` says *Malformed entry*, the line was split or pasted with Windows line endings: run the `echo` line again. This guide runs Docker with `sudo`. Adding `itadmin` to the `docker` group would save typing, but that group is equivalent to root.

2.4 Firewall

Only SSH from your own addresses, the web ports, and RADIUS from the school are let in. Allow SSH **before** turning the firewall on.

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow from YOUR_IP to any port 22 proto tcp
sudo ufw allow 80,443/tcp
sudo ufw allow from SCHOOL_IP to any port 1812:1813 proto udp
sudo ufw enable
sudo ufw status verbose
```

- Repeat the SSH line for each address you manage the server from. If your own IP changes often, use `sudo ufw allow 22/tcp` instead: with key-only SSH, that's still safe.
- Ports 80 and 443 are open to everyone: phones load the sign-in page through them, and Let's Encrypt needs them for certificates.
- **SCHOOL_IP** is the address RADIUS from the vSZ arrives from (see 4.11). Part 8 shows how to read it from the logs if you're unsure.
- The server only *connects out* to the vSZ (API and NBI), so no rule is needed here for those.
- All three containers use the host's network directly, without published ports, so these `ufw` rules apply to them as normal.

2.5 Get the code onto the server

The code is public on GitHub at `github.com/pha-llotter/wifi-platform`, so the server can download it without a key or password. It lives in `/opt/wifi-portal`.

ON THE SERVER

```
sudo mkdir /opt/wifi-portal && sudo chown $USER: /opt/wifi-portal
git clone https://github.com/pha-llotter/wifi-platform.git /opt/wifi-portal
```

The repo holds only code and examples. Every real setting and secret lives in `/etc/wifi-portal` on the server and never goes into Git. Keep your own copies in a folder that Git ignores, such as `server-config/` in your copy of the repo.

2.6 Settings folders and the two secrets

```
sudo mkdir -p /etc/wifi-portal/google-ldap /var/log/wifi-portal /var/lib/wifi-portal
sudo cp /opt/wifi-portal/config.example.json /etc/wifi-portal/config.json
sudo cp /opt/wifi-portal/deploy/freeradius/radius.env.example /etc/wifi-portal/radius.env
sudo chown 1000 /var/log/wifi-portal /var/lib/wifi-portal
openssl rand -hex 32     # 1: the RADIUS shared secret (vSZ ⇔ FreeRADIUS)
openssl rand -hex 32     # 2: the platform token (FreeRADIUS ⇔ platform)
```

Save both values in your password manager, labelled. The first goes on the vSZ (part 4) and in `radius.env`. The second goes in `radius.env` and `config.json` (part 5).

> **Stop here for now**
> Don't start the containers yet. The platform needs the Google and vSZ details first. Carry on with part 3.

PART 3

Google Workspace

Education Fundamentals includes everything here. You'll end up with four things for the server: the Secure LDAP certificate, key and access credentials; an OAuth client for the dashboard sign-in; and a service account key for the directory, alerts and backups.

3.1 Secure LDAP client: checks passwords

1. In the **Admin console** go to **Apps** → **LDAP** → **Add client**. Name it `WiFi RADIUS`.
2. Access permissions, both for the **whole domain** (or every org unit that should use the WiFi):
   - **Verify user credentials**
   - **Read user information**
3. **Download the certificate.** The zip holds a `.crt` and a `.key` file. Rename them `ldap-client.crt` and `ldap-client.key`.
4. Set the client's **Service status** to **On**. It can take a few minutes, and sometimes longer, before Google accepts the certificate.
5. Note the certificate's **expiry date**, shown on the client's page, and put a reminder in the calendar a month before it.
6. **Access credentials** → **Generate new credentials.** Save the username and password: Google shows the password once. Some tenants, including phahs.org.za, let a client with only the certificate see the org units but no users, so every sign-in would fail without them.

3.2 OAuth client: admin sign-in to the dashboard

1. In **Google Cloud console** create a project, for example `nexus-wifi`.
2. **Google Auth Platform** → **Audience**: set it to **Internal**, so only school accounts can sign in.
3. **Clients** → **Create client**: type **Web application**, authorised redirect URI `https://wifi.phahs.org.za/admin/oauth/callback`.
4. Save the **client ID** and **client secret** in the password manager.

This is only for admins signing in to the dashboard. The WiFi sign-in page doesn't use a "Sign in with Google" button, because Google blocks its own sign-in page inside the window phones open for WiFi sign-in. It uses a plain email and password form, checked by Secure LDAP.

3.3 Service account: directory, alerts and backups

1. In the same Cloud project, enable the **Admin SDK API**. Enable the **Gmail API** for alert emails and the **Google Drive API** for off-site backups.
2. **IAM & Admin** → **Service accounts** → **Create**, with no roles. Open it, then **Keys** → **Add key** → **JSON**. Save the file as `service-account.json`.
3. Copy the service account's **client ID** (a long number).
4. In the **Admin console**: **Security** → **Access and data control** → **API controls** → **Domain-wide delegation** → **Add new**. Paste the client ID, and these scopes, comma-separated:

```
https://www.googleapis.com/auth/admin.directory.user.readonly,https://www.googleapis.com/auth/gmail.send,https://www.googleapis.com/auth/drive.file
```

- `admin.directory.user.readonly` reads org units and whether an account is suspended. It's needed.
- `gmail.send` sends alert emails. Leave it out if you don't want alerts.
- `drive.file` uploads nightly backups. It can only see files it uploaded itself. Leave it out if you don't want Drive backups.

> **Your organisation may block service account keys**
> If Google Cloud refuses to make a JSON key, the organisation policy `iam.disableServiceAccountKeyCreation` is on. A Cloud organisation admin can turn it off for this one project.

3.4 Alert mailbox and backup folder (optional)

- **Alerts** are sent as a real Workspace user, for example `wifi-alerts@phahs.org.za`. The service account can send as that user but can't read its mail.

- **Backups**: make a **Shared Drive**, for example `WiFi backups`. Add the alert mailbox user as **Content manager**, make a folder in it, and copy the last part of the folder's URL. That's the folder ID.

3.5 Check the org units

In **Directory** → **Organizational units**, check the exact paths for staff and learners, for example `/Staff`, `/IT Staff` and `/Students`, and fill them in on the worksheet. A rule matches its org unit and everything under it, so every org unit with people who should use the WiFi needs a rule, or a parent with one.

3.6 Copy the files to the server

ON YOUR PC (POWERSHELL), IN THE FOLDER WITH THE THREE FILES

```
scp ldap-client.crt ldap-client.key service-account.json itadmin@SERVER_IP:~
```

ON THE SERVER

```
sudo mv ~/ldap-client.crt ~/ldap-client.key /etc/wifi-portal/google-ldap/
sudo mv ~/service-account.json /etc/wifi-portal/
```

Keep your copies only in the password manager or a Git-ignored folder. Part 5 sets the permissions.

PART 4

Ruckus vSZ

You'll point two RADIUS services at the server, turn on the portal interface, make the sign-in portal, then create the three networks: Nexus Students, Nexus Staff and Guest. Last, you'll open two ports on the school firewall, for the server only.

> **Menu names**
> The paths below are for **SmartZone 6.1** (Essentials). Other versions move some pages. If a menu isn't where this says, type its name into the **search menu** box at the top right of the vSZ. The field names are the same.

4.1 Before you start

- The staff, learner and guest VLANs are tagged on every access point's switch port, and each has a DHCP scope and working DNS.
- You have the server's public IP and the **RADIUS shared secret** from 2.6.

Use **proxy** mode, where the controller sends RADIUS for the APs, for both services. All RADIUS then comes from the vSZ, so FreeRADIUS only has to trust one address, and the APs' own addresses don't matter.

4.2 RADIUS authentication service

Services → Authentication → Proxy (SZ Authenticator) → Create

Field	Value
Name	`Nexus RADIUS`
Service protocol	RADIUS
Primary server: IP	The server's public IP
Port	`1812`
Shared secret	The RADIUS shared secret from 2.6
Secondary server	Leave off

> **The vSZ's "Test AAA" button will say it failed**
> That test sends a plain username and password. The server only accepts a one-time ticket from the sign-in page, or a remembered device, on purpose. Test with a real phone instead (5.8).

4.3 RADIUS accounting service

Services → Accounting → Proxy → Create. Name it `Nexus accounting`, with the same server IP, port `1813` and the same shared secret. This is what feeds the **Usage** page.

4.4 Northbound Portal Interface (NBI)

This is how the platform tells the vSZ "this device may go online". Search the menu for `Northbound`. In 6.1 it's under **Administration → System**.

Field	Value
Enable	Ticked
User name	`wifiplatform`. It goes in `portal.nbi.username`.
Password	Long and random, at least 8 characters with a letter, a number and a special character such as `-`. It goes in `portal.nbi.password`.

The NBI listens on the vSZ's TCP port **9443**, with the same certificate as the web console. Older SmartZone versions ask for the password only: then leave `portal.nbi.username` out.

4.5 The sign-in portal

Services → **Hotspots & Portals** → **Hotspot (WISPr)**, select the school's zone, then **Create**.

Field	Value
Name	`Nexus sign-in`
Smart Client Support	None
Logon URL	**External**. Redirect unauthenticated users to `https://wifi.phahs.org.za/`
Redirected MAC format	Leave the default
Start page	**Redirect to the URL that the user intends to visit**
Session timeout	`720` minutes (12 hours)
Grace period	`60` minutes
Walled garden	`wifi.phahs.org.za`, and nothing else
HTTPS redirection	Leave the default

> **Keep the walled garden to the one name**
> Don't add Apple's or Google's "captive check" addresses, such as `captive.apple.com` or `connectivitycheck.gstatic.com`. If phones can reach those, they think they're already online and never open the sign-in page.

4.6 WLAN: Nexus Students

Network → **Wireless** → **Wireless LANs**, select the school's zone, **Create**.

Setting	Value
Name and SSID	`Nexus Students`. It must match `portal.networks` in config.json **exactly**, including capitals and the space.
Authentication type	**Hotspot (WISPr)**
Method	Open
Encryption	**None**. People need no WiFi password.
Hotspot (WISPr) portal	`Nexus sign-in`
Bypass CNA	**Off.** Phones must open the sign-in page by themselves.
Authentication service	Tick **Use the controller as proxy**, choose `Nexus RADIUS`
Accounting service	Tick **Use the controller as proxy**, choose `Nexus accounting`, interim updates every **10 minutes**
MAC authentication bypass (no redirection)	**On**, MAC format left at the default. This lets remembered devices reconnect without the page.
Access VLAN	The learner VLAN, for example `10`. The same number as in `portal.networks`.
Dynamic VLAN (AAA override)	Off. Each network has its own VLAN.
Advanced → RADIUS options → Called Station ID	**AP MAC**. The vSZ then also sends the network's name, so the server knows which network a device is on.
802.11r fast roaming	Off
Client isolation	On, if you like
WLAN group	Add it to the WLAN group the school's APs use

Can't find MAC authentication bypass?
Look in the Hotspot (WISPr) part of the WLAN and under **Advanced Options**, or search the page for "MAC". If your version doesn't have it, carry on: sign-ins work the same, but devices see the sign-in page again each time they rejoin. Tell IT support, so the platform can remember devices another way.

4.7 WLAN: Nexus Staff

In **Wireless LANs**, select **Nexus Students** and click **Clone**. If your version has no Clone, click **Create** and enter the same settings as 4.6. Then change two things:

- **Name and SSID:** `Nexus Staff`
- **Access VLAN:** the staff VLAN, for example `1`

Check that **Called Station ID** is still **AP MAC** on the copy.

4.8 Check nothing hides the sign-in page

Services → Hotspots & Portals → Portal Detection & Suppression. If a profile there is attached to the school's zone or to either Nexus WLAN, take it off. It can stop phones from showing the sign-in page.

4.9 WLAN: Guest (visitors)

Setting	Value
Name and SSID	`Guest`. It must match `dpsk.ssid` in config.json.
Authentication type, method	Standard usage, **Open**
Encryption	WPA2, AES, with **Dynamic PSK** on (internal DPSK)
Passphrase	The vSZ asks for one even with Dynamic PSK. Use a long random one and keep it in the password manager. Guests never use it: they get their own keys.
DPSK type and length	**Keyboard-Friendly**, **12 to 16** characters, so a key can be typed off a printed slip
DPSK expiration	**Unlimited.** The platform ends guest keys itself, on the day the admin chose.
Access VLAN	The guest VLAN, for example `60`
Client isolation	On, so visitors' devices can't see each other
Accounting service	`Nexus accounting`, if the vSZ offers it on a DPSK WLAN. Guests then show on the Usage page under their key's name.

Make guest keys in the dashboard, not in the vSZ. Keys that already exist on the vSZ can be brought in with **Settings → Adopt keys**.

4.10 API account for guest keys

1. **Administration → Admins and Roles → Roles → Create.** Limit it to the school's zone, with these permissions:
   - **WLAN** and **User/Device/App**: Full access. DPSK keys belong to a WLAN.
   - **SZ**, **AP** and **ICX**: Read.
   - **Admin**: none.
2. Create an administrator `wifi-portal-api` with that role and a long random password. Save the password in the password manager.
3. Don't use a super admin account. If the WLAN list in 5.4 is refused with HTTP 403, the role is too narrow: widen it one step and try again.
4. Open the vSZ web console, click the padlock in the browser, and note the certificate's **SHA-256 fingerprint**. You'll compare it in 5.4.

4.11 School firewall

The server needs to reach two ports on the vSZ, through the school's public IP. Forward them from the server's IP **only**.

Direction	From	To	Port	For
Out	vSZ	Server IP	UDP 1812, 1813	RADIUS: sign-ins, reconnects, usage
In (port forward)	**Server IP only**	vSZ, TCP 8443	Outside 9443 (example)	The vSZ API: guest keys. `dpsk.port`
In (port forward)	**Server IP only**	vSZ, TCP 9443	Outside 9444 (example)	The NBI: letting devices on. `portal.nbi.port`

The outside ports can be anything free. Many firewalls, including FortiGate, use 8443 for their own admin page, which is why the examples use 9443 and 9444.

On a FortiGate

1. **Policy & Objects** → **Addresses** → **Create**: `wifi-server`, type Subnet, the server's IP with `/32`.
2. **Policy & Objects** → **Virtual IPs** → **Create**, twice:
   - `vsz-api`: the WAN port with the school's public IP (or `any`), external IP the school's public IP, map to the vSZ's IP, **Port forwarding** on, TCP, external port `9443` → mapped port `8443`. Under **Optional filters**, set the source to the server's IP.
   - `vsz-nbi`: the same, external port `9444` → mapped port `9443`.
3. **Policy & Objects** → **Services** → **Create**: `VSZ-API` (TCP 8443) and `VSZ-NBI` (TCP 9443).
4. **Policy & Objects** → **Firewall Policy** → **Create**: name `WiFi server to vSZ API`. Fill it in as follows, then drag it above any deny policy from the WAN to that interface:
   - **Incoming:** the WAN port, or the **SD-WAN zone** if the school uses SD-WAN.
   - **Outgoing:** the vSZ's interface.
   - **Source:** `wifi-server`. **Destination:** both Virtual IPs.
   - **Service:** `VSZ-API` and `VSZ-NBI`. **Action:** Accept.
   - **NAT** off, and security profiles off.

> **Never open these ports to the whole internet**
> They're the vSZ's admin API and portal interface. Limit both forwards to the server's IP.

Where RADIUS comes from. With proxy mode, RADIUS reaches the server from the public IP the **vSZ's** traffic leaves on, after the school's NAT. That's `VSZ_ADDRESS` in part 5, and the address in the `ufw` rule in 2.4. With SD-WAN and two ISPs, check it from a PC on the vSZ's network at `https://ifconfig.me`, and make sure the vSZ's traffic always leaves on that ISP.

PART 5

Finish the server and test

Fill in the two settings files, read the vSZ's certificate fingerprint and IDs, start everything, check every connection, and sign in with a real phone.

5.1 Permissions

The platform runs as uid 1000 and reads config.json, the service account key and the LDAP certificate. Only Docker, as root, reads `radius.env`.

```
cd /etc/wifi-portal
sudo chown root:1000 config.json service-account.json google-ldap google-ldap/*
sudo chmod 640 config.json service-account.json google-ldap/*
sudo chmod 750 google-ldap
sudo chown root:root radius.env && sudo chmod 600 radius.env
```

After copying a new config.json with `scp`, run the first two lines again for it, or the platform can't read it.

5.2 config.json: the platform

Edit it with `sudo nano /etc/wifi-portal/config.json`.

Setting	What to put
`publicUrl`	`https://wifi.phahs.org.za`
`google.clientId`, `clientSecret`	From the OAuth client (3.2)
`google.allowedDomains`	`["phahs.org.za"]`. Personal Gmail and other domains are always turned away.
`google.directory.adminEmail`	A Workspace admin the service account acts as, to read the directory
`google.ldap.identity`, `password`	The LDAP client's access credentials (3.1). The sign-in page uses them to check passwords.
`radius.apiToken`	The **platform token** from 2.6
`policy.rules`	Org units to VLANs, from the worksheet, for example `/Staff` and `/IT Staff` → 1, `/Students` → 10. They can be changed later on the dashboard's Rules page.
`portal.networks`	Each sign-in network's exact SSID and its VLAN: `{ "Nexus Students": 10, "Nexus Staff": 1 }`. A person may only use the network whose VLAN their rule gives them.
`portal.rememberDays`	How long a device is remembered: `90` for a term. `0` means sign in every time.
`portal.nbi.host`, `port`	The school's public IP, and the outside port for the NBI (4.11), for example `9444`
`portal.nbi.username`, `password`	From 4.4
`portal.nbi.tlsFingerprint`	Filled in by 5.4
`dpsk.host`, `port`	The school's public IP, and the outside port for the vSZ API (4.11), for example `9443`
`dpsk.username`, `password`	The vSZ API account (4.10)
`dpsk.tlsFingerprint`, `zoneId`, `wlanId`	Filled in by 5.4
`dpsk.guestVlanId`	The guest VLAN, for example `60`
`alerts.from`, `alerts.to`	The alert mailbox, and who gets alerts. Delete the whole `alerts` block to turn alerts off.
`backup.drive.folderId`	The Drive folder ID (3.4). Delete the `drive` block if you don't use Drive backups.
`branding.name`	The name on the sign-in page and guides, for example `Nexus WiFi`
`admin.emails`	Who can sign in to the dashboard

Setting	What to put
`admin.allowFrom`	Optional: IPs or ranges the dashboard accepts, for example the school's public IP and yours. Empty means anywhere, with Google sign-in. The sign-in page is never limited.
`retentionDays`	How long sign-ins and usage are kept, for POPIA. 90 by default.

5.3 radius.env: FreeRADIUS

Edit it with `sudo nano /etc/wifi-portal/radius.env`.

Setting	What to put
`VSZ_ADDRESS`	The address RADIUS arrives from (4.11), usually the school's public IP
`RADIUS_SECRET`	The **RADIUS shared secret** from 2.6, the same as on the vSZ
`PLATFORM_TOKEN`	The **platform token** from 2.6, the same as `radius.apiToken`
`DEFAULT_DOMAIN`	`phahs.org.za`
`LDAP_BASE_DN`, `LDAP_IDENTITY`, `LDAP_PASSWORD`	`dc=phahs,dc=org,dc=za`, and the LDAP access credentials (3.1)
`FALLBACK_VLAN`	Leave empty. The sign-in networks don't use it.
`RADIUS_DEBUG`	`0`. Set it to `1` only while testing (8.1).
Everything else	Leave as it is

5.4 The vSZ's fingerprint and IDs

Do these in this order. Reading the WLAN list needs the fingerprint first.

```
cd /opt/wifi-portal
sudo docker compose build
sudo docker compose run --rm wifi-portal node scripts/vsz-fingerprint.js /etc/wifi-portal/config.json
```

Compare the fingerprint with the one you noted in 4.10. If they match, paste it into **both** `dpsk.tlsFingerprint` and `portal.nbi.tlsFingerprint`: the API and the NBI use the same certificate. If they don't match, stop: something between the server and the vSZ isn't the vSZ.

```
sudo docker compose run --rm wifi-portal node scripts/vsz-wlans.js /etc/wifi-portal/config.json
```

This lists each zone and WLAN with its ID. Copy the school zone's `zoneId`, and the **Guest** `wlanId`, into config.json.

5.5 Start everything

> **Start it in the evening**
> FreeRADIUS restarts itself every 24 hours, counted from when it started. The restart takes a few seconds, and sign-ins during it fail. Starting it after school puts that restart at night.

```
cd /opt/wifi-portal
sudo docker compose up -d
sudo docker compose ps                              # three containers
sudo docker compose logs -f freeradius wifi-portal  # Ctrl+C stops watching
```

- On the very first start, Caddy gets both certificates from Let's Encrypt within a minute, and FreeRADIUS waits for them.
- After a few minutes `docker compose ps` should show all three as **healthy**.
- If `wifi-portal` keeps restarting, `sudo docker compose logs wifi-portal` starts with **Invalid config:** and a list of what to fix in config.json.
- After changing config.json later, `sudo docker compose restart wifi-portal`. After changing radius.env, `sudo docker compose up -d freeradius`.

5.6 Check the connections

```
sudo docker compose exec wifi-portal node scripts/check-connections.js /etc/wifi-portal/config.json
```

It tries each connection the platform needs, and sends a test alert email (`--no-email` skips that). Each line says what's wrong if it fails. The dashboard has the same check under **Settings** → **Check connections**.

Line	Good result
Google directory	✓, with the admin it acts as
Google Secure LDAP	✓, with the certificate and access credentials
Alert email	✓, and the test email arrives
vSZ API (guest keys)	✓, pinned
vSZ portal interface (sign-ins)	✓, pinned. It asks about a made-up device, so *Not found* (code 300) is the right answer.
FreeRADIUS	Grey until the first phone signs in

5.7 Sign in to the dashboard

Open `https://wifi.phahs.org.za/admin` and sign in with an account in `admin.emails`. **Overview** → **Health** shows whether FreeRADIUS is calling, and the state of the Google directory, the vSZ, backups and alerts.

Opening `https://wifi.phahs.org.za/` on a PC shows a short page saying which networks to join. The sign-in form only appears when the vSZ sends a phone there.

5.8 The first phone

Leave `RADIUS_DEBUG=1` in radius.env for this test, so the log shows exactly what the vSZ sends. Then:

1. On a phone, forget any old Nexus network, then join **Nexus Staff**. The sign-in page should open by itself within a few seconds.
2. Sign in with a staff account. The page says **You're connected**.
3. In the dashboard, **Activity** shows `signed_in` from the sign-in page, and **People** shows the device and the network.
4. On the phone, check the IP address is from the staff VLAN's DHCP scope.
5. Turn WiFi off and on. The phone should reconnect **without** the sign-in page, and Activity shows `signed_in` from a remembered device.
6. After 10 to 20 minutes, the phone appears under **Usage** → **Online now**.

Then look at the log on the server:

```
cd /opt/wifi-portal
sudo docker compose logs --tail 60 wifi-portal freeradius | grep -v Status-Server
```

If the sign-in is refused or nothing arrives, go to part 8. When it all works, set `RADIUS_DEBUG=0` and run `sudo docker compose up -d freeradius`.

PART 6

Devices and the pilot

There's nothing to set up on any device: no profile, no certificate, no settings. Run the new networks next to the current one for one or two weeks before everyone moves.

6.1 What people do

1. Join the network for their group: learners **Nexus Students**, staff **Nexus Staff**. There's no WiFi password.
2. The sign-in page opens by itself. They type their school email and password. Just the username, such as `t.smith`, works too.
3. Done. The device is remembered for the rest of the term (90 days), and reconnects by itself.

If	Then
The page doesn't open by itself	Open any `http://` website in the browser, for example `http://neverssl.com`.
The page says their account uses the other network	Join that one. Learners can't use Nexus Staff, and staff can't use Nexus Students.
"That email or password is wrong"	It's their Google password. After 5 wrong tries the account must wait 15 minutes.
The page asks again each time	The device changes its "private WiFi address". On iPhone, set **Private Wi-Fi Address** for that network to **Fixed**, not **Rotating**. Android keeps one address per network by default.
They change their Google password	Nothing changes: the device stays remembered. A blocked or suspended account is cut off at its next reconnect.

6.2 The connection guide

The dashboard's **Settings** → **Connection guide** prints a one-page guide for staff and learners, with the networks, the steps above and a privacy note. Check its privacy wording against the school's ICT policy before handing it out.

6.3 Pilot checklist

Run it with IT and a few teachers and learners. Tick each item before telling everyone.

	Check	How
☐	**Every connection works**	**Settings** → **Check connections** is all green, and the test email arrived.
☐	**Passwords**	A staff and a learner account sign in. A wrong password is refused with "That email or password is wrong".
☐	**The right network**	A learner on Nexus Staff is told to use Nexus Students. A staff member on Nexus Students is told to use Nexus Staff.
☐	**The right VLAN**	A staff phone and a learner phone get addresses from their own VLANs.
☐	**Remembered devices**	Turn WiFi off and on, and walk out of range and back: no sign-in page. Activity shows a remembered device.
☐	**Usage arrives**	After 10 to 20 minutes online, the phone shows under **Usage** → **Online now** with growing figures.
☐	**Every kind of device**	iPhone, Android, a Windows laptop, a Mac, a school Chromebook and a personal one. Each should open the page by itself.
☐	**Turning people away**	Block a test account: its devices are forgotten, and it can't sign in again. The reason shows under **Activity**. Unblock it. Do the same with a suspended test account in Google.
☐	**Guests**	Make a guest key, print the slip, join Guest with it, then end it: the phone can't join again.
☐	**Server down**	After school, `sudo docker compose stop wifi-portal`. Phones already online stay online, and new sign-ins fail. Then `sudo docker compose start wifi-portal`.
☐	**Backups**	**Settings** → **Back up now**. With Drive set up, the copy appears in the Shared Drive.

	Check	How
☐	**A busy morning**	During the first full morning, watch **Overview** for `directory_error` and `portal_error`.

6.4 Going live

- Print and hand out the connection guide (6.2).
- Turn off the old school network, or hide it, once everyone has moved.
- Check `RADIUS_DEBUG=0` in radius.env.
- Point an uptime monitor (Uptime Kuma, or your RMM) at `https://wifi.phahs.org.za/healthz`.
- Watch for `directory_error` and `portal_error` alerts in the first busy week.

PART 7

Looking after it

Day to day, the dashboard and the alert emails tell you when something needs attention. This part covers the jobs around them.

7.1 Updating the platform

After school, from `/opt/wifi-portal`:

```
git pull
sudo docker compose up -d --build
sudo docker compose ps
```

Only containers whose code changed are restarted. Connected devices stay connected while the platform restarts. Sign-ins during the restart fail for a few seconds.

7.2 Server updates and reboots

- Security updates install themselves (2.2). Run `sudo apt update && sudo apt full-upgrade` now and then for everything else.
- When `/var/run/reboot-required` exists, reboot after school or in the holidays with `sudo reboot`. All three containers come back by themselves.

7.3 Backups: what's covered and what isn't

What	Where it's kept
The database: sign-ins, people, remembered devices, usage, guest keys, rules	Copied every night to `/var/lib/wifi-portal/backups/` (the last 14), and to the Drive folder if set up (the last 30, gzipped).
Settings and secrets in `/etc/wifi-portal`	**Not backed up automatically.** Make a copy after setup and after each change, as below.
The code	On GitHub: `github.com/pha-llotter/wifi-platform`
Let's Encrypt certificates	Not needed: Caddy gets new ones by itself.

```
sudo tar czf ~/wifi-portal-etc-$(date +%F).tgz -C /etc wifi-portal
sudo chown $USER ~/wifi-portal-etc-*.tgz      # then copy it to your PC with scp
```

That file holds every secret. Store it encrypted, for example as an attachment in the password manager, and delete it from the server.

7.4 Restoring the database

```
cd /opt/wifi-portal
sudo docker compose stop wifi-portal
sudo mkdir -p /var/lib/wifi-portal/before-restore
sudo mv /var/lib/wifi-portal/wifi-portal.db* /var/lib/wifi-portal/before-restore/
sudo cp /var/lib/wifi-portal/backups/wifi-portal-2026-11-20-00-00-00.db /var/lib/wifi-portal/wifi-portal.db
sudo chown 1000 /var/lib/wifi-portal/wifi-portal.db
sudo docker compose start wifi-portal
```

- Moving `wifi-portal.db*` aside also moves the database's `-wal` and `-shm` files. Left in place, they would be mixed into the restored copy.
- For a copy from Drive, run `gunzip` on it first.
- Then use **Settings** → **Adopt keys** to bring back guest keys made since that copy. Devices remembered since that copy sign in once more.

7.5 Rebuilding a lost server

1. Do 2.1 to 2.5 on a new server. If the IP changed, update both DNS records, the vSZ RADIUS services, the two port forwards' source filter and the `ufw` rules.
2. Put back `/etc/wifi-portal` from your copy: `sudo tar xzf wifi-portal-etc-DATE.tgz -C /etc`. Skip 2.6: it would overwrite the restored settings with the examples.
3. Make the data folders:
   `sudo mkdir -p /var/log/wifi-portal /var/lib/wifi-portal && sudo chown 1000 /var/log/wifi-portal /var/lib/wifi-portal`.
   Then restore the latest database (7.4) into `/var/lib/wifi-portal`.
4. Start as in 5.5, then check as in 5.6 and 5.8.

7.6 Things that expire or change

What	What to do
Google Secure LDAP certificate	Before it expires, generate a new one on the LDAP client, copy it to `/etc/wifi-portal/google-ldap/` with the same names and permissions, then `sudo docker compose restart wifi-portal freeradius` after school.
The vSZ's own certificate	If it's ever replaced, sign-ins and guest keys stop with a fingerprint error. Run 5.4 again, compare, and update both fingerprints.
The NBI password	Change it on the vSZ (4.4) and in `portal.nbi.password` together, then `sudo docker compose restart wifi-portal`.
A network's name or VLAN	Change the WLAN and `portal.networks` together. The SSID must match exactly.
Let's Encrypt	Automatic. Caddy renews, and FreeRADIUS picks it up at its daily restart.
IT staff changes	Update `admin.emails` and `alerts.to`, then `sudo docker compose restart wifi-portal`.
Changing the RADIUS secret	Change it on both vSZ services and in `radius.env` at the same time, after school, then `sudo docker compose up -d freeradius`.
Changing the platform token	Change `radius.apiToken` and `PLATFORM_TOKEN` together, then `sudo docker compose up -d --force-recreate`.

7.7 Logs

```
sudo docker compose logs --tail 200 freeradius      # RADIUS: tickets, reconnects, usage
sudo docker compose logs --tail 200 wifi-portal     # the platform and the sign-in page
sudo tail -f /var/log/wifi-portal/wifi.jsonl        # every sign-in and admin action
```

Docker keeps 5 × 20 MB of output per container. `wifi.jsonl` rotates at 50 MB and keeps 5 old copies.

PART 8

When something doesn't work

Start with the dashboard: **Activity** gives the reason for every refusal, and **Overview** → **Health** shows which connection is down. The table below covers the rest.

8.1 Seeing every request in detail

```
sudo nano /etc/wifi-portal/radius.env      # set RADIUS_DEBUG=1
sudo docker compose up -d freeradius
sudo docker compose logs -f freeradius     # try a sign-in, then Ctrl+C
```

What debug output shows

Each RADIUS request in full: device addresses, the network name in *Called-Station-Id*, and the one-time tickets. Passwords never reach FreeRADIUS, so they never appear. It's noisy, so set `RADIUS_DEBUG=0` again when you're done and run `sudo docker compose up -d freeradius`.

8.2 Symptoms

You see	Likely cause	What to do
The phone joins, but no sign-in page opens	Bypass CNA is on, a suppression profile is attached, or the walled garden has a captive-check address	Check 4.5, 4.6 and 4.8. For now, open `http://neverssl.com` on the phone.
The browser can't load the sign-in page	`wifi.phahs.org.za` isn't in the walled garden, or DNS doesn't work on that VLAN	Check the walled garden (4.5), and the VLAN's DHCP and DNS.
"This page has expired"	The page was left open for more than 30 minutes	Turn WiFi off and on, and the page opens again.
"That email or password is wrong", with the right password	Secure LDAP: the client is off, a permission is missing, or the access credentials are wrong	Check 3.1. Run **Check connections**: its LDAP line says what Google answered.
"The school WiFi can't check passwords right now"	The platform can't reach Secure LDAP	**Check connections**. A TLS error usually means the server's clock is wrong (`timedatectl`) or the certificate files don't match.
"Your account uses the ... network"	Working as intended: the rule's VLAN belongs to the other network	Join the other network. If the person is in the wrong org unit, fix it in Google, or change the rule.
"Your password is right, but the WiFi didn't respond"	The server can't reach the NBI	**Check connections**. Check the NBI is enabled (4.4) and the NBI port forward (4.11).
"Your password is right, but the WiFi didn't let this device on"	The vSZ's RADIUS check of the ticket failed	Look in the FreeRADIUS log (8.1). Usually one of the next two rows. Activity may say *not signed in on the portal*.
FreeRADIUS log: *Ignoring request ... from unknown client 198.51.100.7*	RADIUS arrives from a different address than `VSZ_ADDRESS`	Set `VSZ_ADDRESS` to the address in the message, fix the `ufw` rule to match, then `sudo docker compose up -d freeradius`.
Nothing in the FreeRADIUS log at all	A firewall in the way: the server's, the provider's, or the school's	On the server, `sudo tcpdump -ni any udp port 1812` while a phone tries. No packets means it's being blocked before the server.
Every sign-in fails after the password; Health says FreeRADIUS hasn't called	The platform token differs between config.json and radius.env	Make `radius.apiToken` and `PLATFORM_TOKEN` the same, then `sudo docker compose up -d --force-recreate`.
A remembered phone sees the sign-in page every time	MAC authentication bypass is off; Called Station ID doesn't include the network name (Activity shows `portal_ssid_error`); or the phone's private address rotates	Check 4.6 on both WLANs, and 6.1.

You see	Likely cause	What to do
Signed in, but on the wrong VLAN	The WLAN's Access VLAN doesn't match `portal.networks`	Make them the same (4.6, 4.7, 5.2).
Signed in, but the device gets no IP address	That VLAN doesn't reach the AP, or has no DHCP scope	Check the switch port is tagged for the VLAN, and the DHCP scope.
The vSZ's "Test AAA" fails	Expected: it isn't a sign-in from the page	Test with a phone (5.8).
Refused: rule, blocked, suspended or wrong domain	Working as intended	Activity gives the reason. Change the rule, the block or the Google account.
`wifi-portal` keeps restarting	A mistake in config.json, or it can't read the file	`sudo docker compose logs wifi-portal` lists each problem under *Invalid config*. Check the permissions (5.1).
"doesn't match the pinned fingerprint"	The vSZ certificate changed, or something is intercepting	Run 5.4 again and compare with the vSZ console before updating.
Guest keys: sign-in failed or HTTP 403	The API account's password or role	Check the password, then widen the role (4.10).
NBI: *Wrong request password* (code 306)	The NBI user name or password differs from the vSZ	Set them again on both sides (4.4, 5.2).
Can't reach the vSZ API or NBI	The school's port forwards	Check 4.11, including that they allow the server's IP.
`directory_error` alerts	Google directory or quota trouble, or delegation scopes	Check the delegation entry (3.3).

APPENDIX

Ports and files

Ports

On	Port	Open to	For
Server, in	TCP 22	Your IPs	SSH
Server, in	TCP 80, 443	Anywhere	The sign-in page, the dashboard, Let's Encrypt
Server, in	UDP 1812, 1813	The RADIUS source IP only	Sign-ins, reconnects and usage from the vSZ
Server, out	TCP 636	ldap.google.com	Password checks
Server, out	TCP 443	Google APIs, Let's Encrypt, Docker Hub, GitHub	Directory, alerts, backups, certificates, updates
Server, out	TCP 9443, 9444 (examples)	The school's public IP	The vSZ API (guest keys) and the NBI (sign-ins)
Server, local only	TCP 8080; UDP 18120, 18121	127.0.0.1 only: never open	Platform API; FreeRADIUS internals and health check
School, out	UDP 1812, 1813	The server	RADIUS from the vSZ
School, in (port forward)	9443 → vSZ 8443, 9444 → vSZ 9443	The server's IP only	The vSZ API and the NBI
Sign-in networks, walled garden	TCP 443	`wifi.phahs.org.za` only	The sign-in page before a device is let on

Files on the server

Path	What it is
`/opt/wifi-portal`	The code. `docker compose` commands run from here.
`/etc/wifi-portal/config.json`	Platform settings (root:1000, 640)
`/etc/wifi-portal/radius.env`	FreeRADIUS settings and secrets (root, 600)
`/etc/wifi-portal/service-account.json`	Google service account key (root:1000, 640)
`/etc/wifi-portal/google-ldap/ldap-client.crt`, `.key`	Google Secure LDAP client certificate (root:1000, 640)
`/var/lib/wifi-portal/wifi-portal.db`	The database, including remembered devices
`/var/lib/wifi-portal/backups/`	Nightly copies, the last 14
`/var/log/wifi-portal/wifi.jsonl`	Sign-ins and admin actions, one JSON line each
Docker volume `caddy-data`	Let's Encrypt certificates

Containers

Name	Does
`caddy`	HTTPS for the sign-in page and the dashboard
`wifi-portal`	The platform: the sign-in page, password checks, the hand-over to the vSZ, remembered devices, the dashboard, guest keys, usage, alerts, backups
`freeradius`	Answers the vSZ's RADIUS: checks one-time tickets and remembered devices with the platform, and passes on usage